Interpretability-Driven Dermatological Diagnosis via Multimodal Foundation Models and Quantifiable Severity Index (QSI)

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Bhargav R Rao
School of Computer Science and Engineering
RV University
Bengaluru, India
bhargavvraumtech25@rvu.edu.in

Dr. Karthikeyan Periyasami
Department of Computer Science
RV University
Bengaluru, India
karthikeyanp@rvu.edu.in

***Abstract*—A correct diagnosis of skin conditions is inherently multi-modal and depends on the seamless fusion of highly accurate visual morphologies with clinical data. We propose a new diagnostic scheme, named Dino-Derm, where the deep features of DINOv2 Vision Transformer (ViT) are utilized as the feature extractor and are fused with patient-level information: Age, Sex, and Anatomical Location through a late fusion-based neural network framework. Given the black-box nature of deep models in the clinical setting, we propose the Quantifiable Severity Index (QSI) using Attention Rollout as the metric to quantify the severity of the skin condition mathematically by leveraging pathologically significant visual features. Our experimental results show that by employing multimodal fusion, we achieve an average accuracy rate of 82% for HAM10000. Through a mathematical framework, Dino-Derm bridges the gaps between the representation obtained from the foundation model and the clinical prior knowledge.**

***Index Terms*—Dermoscopy, DINOv2, Multimodal Fusion, Explainable AI (XAI), Attention Rollout, Clinical Decision Support.**

I. Introduction

One of the most common cancers around the world is skin cancer, and melanoma specifically is one of the deadliest skin cancers, making early detection crucial. Skin cancer diagnosis using dermoscopy techniques plays an important role in skin cancer treatment. However, this process faces many challenges since there is very high similarity and variability among different types of skin lesions. Such issues can lead to false diagnoses of skin cancer when resources are lacking.

Deep learning techniques like CNNs and ViTs have shown great promise in automatically classifying skin lesions. The reason behind this is that the algorithms can achieve a higher classification performance. However, such a process suffers from two major problems. First, most of these algorithms only consider images as inputs but do not include any information about the metadata of patients like age and gender. Furthermore, such algorithms work like black boxes, as they make predictions without explaining their reasoning.

In order to alleviate these issues, the study presents Dino-Derm, a new diagnosis framework based on interpretability and multimodal learning in which visual features extracted from a self-supervised foundation model are fused with patient metadata. DINOv2, a cutting-edge Vision Transformer pretrained with self-supervised learning algorithms, is used for capturing lesion features. They are later merged with patient metadata in an effective manner using late-fusion architecture, leading to better decisions.

On top of improved performance and higher accuracy in diagnosis, Dino-Derm introduces a new evaluation metric called Quantifiable Severity Index (QSI). Contrary to common methods like Grad-CAM, QSI allows quantification of lesion abnormality by inspecting attention maps and feature contrast of images. In other words, it allows a more quantitative interpretation of results as opposed to mere visual explanation, helping bridge artificial intelligence and medicine.

Dino-Derm is tested on the HAM10000 dataset. Better robustness and balanced performance are observed compared to other methods.

The main contributions of this paper are summarized as follows:

1) A multimodal diagnostic framework combining DINOv2-based visual features with clinical metadata.
2) A novel Quantifiable Severity Index (QSI) for interpretable and measurable explanation of model predictions.
3) A late-fusion architecture that enhances classification robustness across diverse skin lesion categories.
4) Comprehensive evaluation on the HAM10000 dataset with statistical validation of interpretability scores.

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II. Related Work

A. Vision Transformers and Foundation Models

The recent developments in computer vision are mainly attributed to Vision Transformers (ViTs). These have been shown to outperform the conventional Convolutional Neural Networks (CNNs) on various image classification tasks. The ViT is an architecture that exploits self-attention mechanisms to capture global dependencies. In another work, a self-supervised framework called DINO was introduced as a means to achieve enhanced representation learning without relying on labelled datasets. Very recently, Oquab et al. proposed DINOv2 as a foundation model for learning effective visual representations. This approach differs from CNN-based models because it focuses on capturing detailed structural properties, which make it useful in medical imaging tasks.

B. Multimodal Learning in Dermatology

Multiple works have considered the use of clinical metadata with dermoscopic images for better diagnoses. For example, Yap et al. illustrated that the inclusion of patient-related metadata, including age and lesion localization, can contribute to the effectiveness of classifiers when paired with images. Another example is from Huang et al., who introduced a set of multimodal learning models used in medical imaging processing, emphasizing the role of clinical contexts. Nevertheless, all these works utilize traditional CNN architectures and do not utilize foundation models. Additionally, most current fusion methodologies fail to effectively capture inter-modal correlations. In contrast, we propose a late-fusion framework based on the visual embeddings of DINOv2 and patient metadata.

C. Explainable AI in Medical Imaging

Interpretability is one of the main obstacles to using deep learning in the medical field. Methods like Grad-CAM offer qualitative insights by marking regions in the input image that are crucial for the prediction; however, they are mostly non-mathematical in nature. Some recent research has focused on transformer explainability with attention rollout techniques proposed by Abnar and Zuidema. These techniques help in understanding how information propagates through each layer of the transformer network; however, none of these methods can give a measure of severity of the lesion. To fill this gap, we propose a new measure called Quantifiable Severity Index (QSI).

III. Proposed Methodology

A. Multimodal Feature Fusion Architecture

The proposed Dino-Derm framework adopts a dual-branch multimodal architecture to process heterogeneous medical data by combining dermoscopic image features with structured clinical metadata.

Visual Branch: A pre-trained DINOv2 Vision Transformer is employed as the primary feature extractor. The model is initialized with self-supervised weights and its backbone parameters are frozen during training to preserve robust feature representations. For an input image x, a 768-dimensional feature vector is extracted from the [CLS] token:

$$f_v = \text{DINOv2}(x), \quad f_v \in \mathbb{R}^{768} \tag{1}$$

Metadata Branch: Clinical metadata, including age, biological sex, and anatomical localization, are processed through a Multi-Layer Perceptron (MLP). After normalization and encoding, the metadata vector m is projected into a lower-dimensional embedding:

$$f_m = \phi(m), \quad f_m \in \mathbb{R}^{32} \tag{2}$$

where $\phi(\cdot)$ denotes the MLP transformation.

Late Fusion Strategy: The visual and metadata embeddings are concatenated to form a unified feature representation:

$$f_{\text{fusion}} = [f_v \oplus f_m] \tag{3}$$

This fused vector is passed through a classification head to predict the probability distribution over C classes:

$$\hat{y} = \text{Softmax}(W f_{\text{fusion}} + b) \tag{4}$$

B. Quantifiable Severity Index (QSI)

To improve interpretability, we propose the Quantifiable Severity Index (QSI), a novel metric designed to quantify the pathological contrast of a lesion using transformer attention mechanisms.

Attention Extraction: Let $A \in \mathbb{R}^{N \times N}$ represent the attention matrix from the final transformer layer. The attention weights from the [CLS] token to all spatial patches are extracted and averaged across heads to obtain an importance map:

$$A_{\text{cls}} = \frac{1}{H} \sum_{h=1}^{H} A_h^{\text{cls}} \tag{5}$$

Feature Separation: Let $\text{feat}_{\text{lesion}}$ denote the mean feature vector of the top-k (e.g., top 20%) most-attended patches, and $\text{feat}_{\text{global}}$ denote the average feature vector across all patches:

$$\text{feat}_{\text{lesion}} = \frac{1}{k} \sum_{i \in \text{Top-}k} z_i \tag{6}$$

$$\text{feat}_{\text{global}} = \frac{1}{N} \sum_{i=1}^{N} z_i \tag{7}$$

QSI Formulation: The QSI is computed as a weighted combination of cosine dissimilarity and the relative area of high-attention regions:

$$QSI = \left(0.7 \cdot \left(1 - \frac{\text{feat}_{\text{lesion}} \cdot \text{feat}_{\text{global}}}{\|\text{feat}_{\text{lesion}}\| \|\text{feat}_{\text{global}}\|}\right) + 0.3 \cdot \frac{\text{Area}_{\text{top-}k}}{\text{Area}_{\text{total}}}\right) \times 100 \tag{8}$$

The QSI score ranges from 0 to 100, where higher values indicate greater deviation from normal skin patterns, providing a quantitative measure of lesion abnormality.

C. Training and Validation

The model is trained and evaluated on the HAM10000 dataset, which consists of dermoscopic images spanning seven diagnostic categories. All images are resized to 518×518 pixels and normalized using ImageNet statistics.

The network is optimized using the AdamW optimizer with a learning rate of 1×10^{-3} and a cross-entropy loss function:

$$\mathcal{L} = -\sum_{c=1}^{C} y_c \log(\hat{y}_c) \tag{9}$$

To ensure training stability and reproducibility, model checkpoints are saved after each epoch, allowing selection of the best-performing model based on training loss convergence.

D. Overall System Architecture

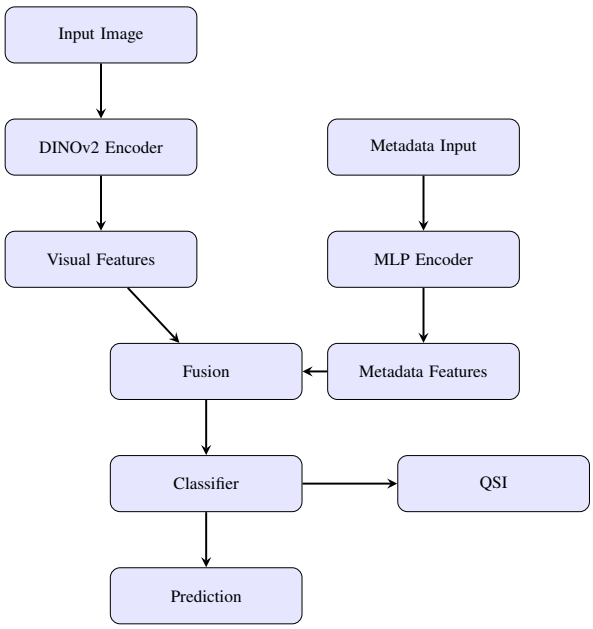

Fig. 1. Overview of the proposed Dino-Derm architecture.

As shown above, the proposed Dino-Derm framework follows a multimodal architecture combining visual features extracted using DINOv2 with clinical metadata for classification and interpretability.

The architecture of the proposed Dino-Derm framework is illustrated in Fig. 1. The system is designed as a multimodal pipeline that integrates dermoscopic image features with structured clinical metadata to improve diagnostic robustness.

The visual branch processes the input dermoscopic image using a pre-trained DINOv2 Vision Transformer. The image is divided into patches and encoded through self-attention layers, producing a 768-dimensional feature vector from the [CLS] token that captures global lesion characteristics such as texture, color, and structural patterns.

In parallel, the metadata branch processes patient-specific information including age, biological sex, and anatomical localization. These features are normalized and passed through a Multi-Layer Perceptron (MLP), resulting in a compact 32-dimensional embedding that represents clinical context.

The outputs of both branches are combined using a late-fusion strategy, where the visual and metadata embeddings are concatenated to form a unified feature representation. This fused vector is then passed through a classification head to predict the probability distribution over seven skin lesion categories.

To enhance interpretability, the framework incorporates a Quantifiable Severity Index (QSI) module. Attention maps extracted from the transformer are used to identify regions contributing to the prediction. The QSI quantifies the pathological deviation of the lesion by measuring feature contrast and attention distribution, providing a clinically meaningful interpretation of model decisions.

IV. Experiments and Results

A. Experimental Setup

The proposed Dino-Derm framework was implemented using PyTorch and trained on a GPU-enabled environment. A pre-trained DINOv2 backbone was used as a frozen feature extractor to ensure stable representation learning. The model was trained for 5 epochs using the AdamW optimizer with a learning rate of 1×10^{-3}.

B. Evaluation Metrics

Model performance was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. A confusion matrix was used for class-wise analysis. Additionally, the Quantifiable Severity Index (QSI) was evaluated to measure interpretability across lesion categories.

C. Classification Performance

The proposed model achieved an overall accuracy of 82% on the HAM10000 dataset. High performance was observed for melanocytic nevi (nv), achieving a recall of 0.96 and an F1-score of 0.93. Vascular lesions (vasc) were classified with a precision of 1.00, demonstrating strong detection capability for rare classes.

However, relatively lower performance was observed for melanoma (mel) and actinic keratoses (akiec), primarily due to class imbalance and visual similarity with other lesion types.

D. Confusion Matrix Analysis

The confusion matrix reveals that misclassifications predominantly occur between visually similar classes. In particular, melanoma (mel) is frequently confused with benign keratosis (bkl) and melanocytic nevi (nv), highlighting the inherent difficulty of distinguishing malignant lesions from benign counterparts.

E. QSI-Based Interpretability Analysis

The Quantifiable Severity Index (QSI) was computed using attention rollout to quantify lesion abnormality. The average QSI score across the dataset was approximately 14, with variations observed across different disease categories.

Classes such as vascular lesions (vasc) and dermatofibroma (df) exhibited higher variability in QSI values, while

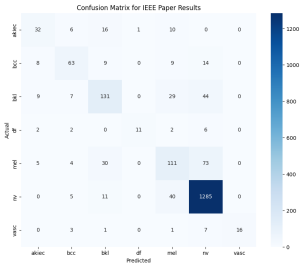

Fig. 1. Confusion matrix illustrating classification performance across lesion categories.

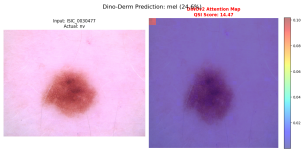

Fig. 2. Visualization of transformer attention using attention rollout. The highlighted regions indicate areas contributing to the model's decision. The attention map provides coarse localization rather than precise segmentation. The corresponding QSI score quantifies lesion abnormality.

Fig. 3. Statistical validation of QSI scores across HAM10000 classes. Error bars represent standard deviation.

melanocytic nevi (nv) showed relatively consistent scores. These results demonstrate the effectiveness of QSI in providing a quantitative measure of lesion severity.Lower QSI values indicate that the lesion features closely resemble global skin patterns, suggesting benign characteristics. In contrast, higher QSI values reflect stronger deviation from normal patterns, indicating potential abnormality. This aligns with clinical intuition, where irregular texture and structure are key indicators of malignancy.

F. Training Stability

Training logs indicate stable convergence, with cross-entropy loss decreasing from approximately 1.18 to 0.43 over 5 epochs, confirming effective optimization and model stability.

V. Discussion

The results demonstrate that the proposed Dino-Derm framework effectively combines multimodal learning with foundation model representations to achieve robust classification performance. The integration of clinical metadata with visual features contributes to improved diagnostic consistency, particularly in common lesion categories such as melanocytic nevi.

However, certain limitations are observed. The model exhibits reduced performance in challenging classes such as melanoma and actinic keratoses, primarily due to class imbalance and high visual similarity between lesion types. Additionally, the attention maps generated from the transformer backbone provide only coarse localization of important regions, which is consistent with the global attention mechanism of Vision Transformers.

To address these limitations, the proposed Quantifiable Severity Index (QSI) plays a critical role by offering a quantitative measure of lesion abnormality. Unlike purely visual explanations, QSI provides a more stable and interpretable metric that complements attention-based visualization.

Overall, the findings suggest that while transformer-based attention improves global understanding, combining it with quantitative interpretability metrics and multimodal inputs leads to a more reliable and clinically meaningful diagnostic system.

VI. Conclusion

This paper presented Dino-Derm, an interpretability-driven multimodal framework for skin lesion classification that integrates visual features from a self-supervised foundation model with structured clinical metadata. By leveraging DINOv2 as a robust feature extractor and incorporating patient-specific information through a late-fusion architecture, the proposed system achieves improved diagnostic robustness across multiple lesion categories.

To address the limitations of black-box deep learning models, this work introduced the Quantifiable Severity Index (QSI), a novel metric designed to provide quantitative interpretability. Unlike traditional visualization techniques, QSI offers a measurable indication of lesion abnormality by combining attention-based feature contrast with spatial distribution.

Experimental results on the HAM10000 dataset demonstrate that the proposed approach achieves strong classification performance while maintaining interpretability. The results highlight the effectiveness of multimodal fusion and the reliability of QSI as a clinically relevant explanation tool.

Overall, Dino-Derm provides a step toward trustworthy and explainable AI systems in dermatology, with potential applications in real-world clinical decision support.

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